BOILEAU GUILLAUME

Ingénieur IVV / Validation Systèmes | Python, Automatisation de Tests, Analyse d'Anomalies & Systèmes Complexes

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EXPERIENCE

Software Engineer - System Integration, Production Diagnostics & Hardware Interfaces

VEV Platform Services France

2025 – 2026 · Valbonne, France

Context: Development and integration of software services for EV charging infrastructure within a CPMS platform connected to operational hardware systems.
Objective: Support robust software integration, operational data flows and interface reliability between platform services and charging station systems.

- **Software development:** Developed web and backend services using TypeScript, Angular and Node.js in an operational industrial environment.
- **System integration:** Contributed to the integration and maintenance of software services interacting with field hardware and charging infrastructure.
- **Hardware/software interfaces:** Analysed interactions between platform services, operational data flows and charging station systems.
- **Operational validation:** Checked data consistency, service behaviour and interface continuity in production-like conditions.
- **Incident analysis:** Investigated production issues involving software behaviour, data exchanges and hardware-connected systems.
- **First-level diagnostics:** Analysed abnormal behaviours, communication issues and data inconsistencies before escalation or correction follow-up.
- **Data exchange:** Implemented and monitored FTP-based operational data exchange services.
- **Agile tooling:** Used Zenhub to support task tracking, backlog follow-up, iterative development and technical coordination.

Senior R&D Engineer - Space Systems, IVV, Test Automation & Validation Workflows

Laboratoire Lagrange, Observatoire de la Côte d'Azur, CNES

2023 – 2025 · Nice, France

Context: Engineering activities for the LISA ESA/NASA space mission, involving complex software chains, model integration, simulation workflows, validation campaigns and international coordination.
Objective: Develop, integrate and validate software and simulation workflows under strong consistency, traceability, performance and verification constraints.

- **Space systems:** Contributed to mission-level analysis workflows for the LISA ESA/NASA space mission in a large international technical collaboration.
- **V-cycle logic:** Supported engineering activities aligned with specification, integration, verification and validation logic for complex mission-level software workflows.
- **IVV logic:** Structured integration, verification and validation workflows for complex models, software chains and simulation outputs.
- **Python automation:** Developed Python tools to automate model comparison, consistency checks, validation workflows and technical analysis tasks.
- **Test strategy:** Defined comparison campaigns, validation checks and acceptance logic for technical outputs produced by multiple contributors.
- **Non-regression logic:** Compared successive outputs, configurations and model behaviours to detect deviations, regressions and unexpected changes.
- **Script execution:** Executed analysis and validation scripts, reviewed outputs and consolidated results for technical interpretation.
- **Anomaly analysis:** Investigated inconsistencies, deviations and unexpected behaviours affecting result reliability and interpretation.
- **Software platform:** Developed CATpyLISA, a Python platform for large-scale model integration, comparison, validation and technical review. GitLab:CATpyLISA
- **Technical reporting:** Produced validation notes, technical documentation, analysis summaries and evidence supporting technical reviews.
- **Technical coordination:** Coordinated contributors, supervised interns and supported validation activities across multiple stakeholders.
- **Documentation tooling:** Used Confluence to support technical documentation, coordination notes, validation follow-up and knowledge sharing within project activities.

Data Scientist - Diagnostics, Test Analysis & Performance Validation

Universiteit Antwerpen, Belgium

2021 – 2023 · Antwerp, Belgium

Context: Data analysis and performance studies for precision instruments in a high-requirement technical environment linked to gravitational-wave detector performance.
Objective: Identify, characterize and mitigate sources affecting system sensitivity using robust statistical, computational and validation-oriented methods.

- **Performance validation:** Analysed sources affecting system sensitivity, performance stability and measurement quality.
- **Technical diagnostics:** Identified contributors to performance degradation and supported prioritisation of mitigation actions.

PROFILE FOR SCALIAN

Systems validation engineer with a strong background in Python automation, IVV logic, simulation workflows, test analysis, anomaly investigation and software/hardware interface understanding. Experience developed through major international collaborations including **LISA**, **LIGO–Virgo–KAGRA** and **Einstein Telescope**, complemented by recent industrial software experience involving operational services connected to hardware infrastructure.
For an **embedded systems validation / IVVQ / Defence** role, I bring a practical engineering mindset: developing and executing Python scripts, analysing test and simulation outputs, identifying anomalies, documenting technical evidence, supporting non-regression logic and coordinating with multidisciplinary teams.

- Strong experience with complex systems under integration, validation, consistency and traceability constraints.
- Development and execution of Python scripts for automation, model comparison, consistency checks and validation workflows.
- Verification and validation of software chains, simulation workflows, model outputs and technical deliverables.
- Analysis of anomalies, interface behaviours, performance degradation and system-level limitations.
- Experience with software/hardware interfaces through operational industrial systems connected to charging infrastructure.
- Familiarity with technical contexts involving protocol constraints, operational data exchanges and interface reliability.
- Structured reporting: validation notes, technical documentation, progress summaries and technical review support.
- Understanding of V-cycle engineering logic: requirements, specification, integration, verification, validation and technical evidence.
- Experience working in Agile software environments with iterative development, task coordination and stakeholder interaction.
- Use of collaborative engineering tools including Confluence for technical documentation and Zenhub for Agile task tracking and coordination.

FIT FOR EMBEDDED SYSTEMS VALIDATION

- Relevant background for **IVVQ**, validation campaigns, non-regression logic and test result analysis.
- Strong practical experience with **Python automation**, scientific computing, analysis pipelines and reproducible workflows.
- Ability to execute scripts, inspect outputs, analyse first-level anomalies and document findings clearly.
- Experience producing technical documentation, validation evidence, analysis reports and review material.
- Comfortable with JIRA-style issue tracking, anomaly follow-up and structured technical communication.
- Strong fit for complex and critical environments requiring rigor, autonomy, analytical thinking and traceability.

LANGUAGES

English ●●●●○

Spanish ●●○○○

EDUCATION

PhD in Physics

Université Côte d'Azur

2018 – 2021 · Nice, France

- **Analysis pipelines:** Developed Python-based pipelines for noise source identification, characterization and mitigation.
- **Test data analysis:** Analysed measurement and simulation outputs to identify abnormal behaviours and possible degradation sources.
- **First-level anomaly analysis:** Investigated unexpected behaviours, dominant contributors, weak points and possible correction or mitigation directions.
- **Validation reporting:** Produced reproducible and traceable technical analyses for international engineering and research stakeholders.
- **System impact:** Contributed to studies identifying environmental and instrumental constraints for next-generation gravitational-wave observatories.

Junior R&D Engineer - Signal Analysis, Simulation & Software Validation

ARTEMIS, Observatoire de la Côte d'Azur

2018 – 2021 | Nice, France

Context: Engineering and analysis activities for gravitational-wave mission preparation, involving signal analysis, simulation, software workflows and noise characterization.
Objective: Develop robust analysis methods for complex signals and support technical investigations in demanding system environments.

- **Signal analysis:** Developed methods for the separation of overlapping low-SNR signals in complex datasets.
- **Simulation workflows:** Built simulation approaches for complex signal environments, uncertainty studies and large-scale datasets.
- **Software validation:** Compared simulated outputs, model behaviours and analysis results to assess consistency and robustness.
- **Technical investigations:** Investigated noise sources through cross-sensor correlation analyses in diagnostic contexts.
- **Mission performance:** Supported the identification of limiting factors affecting mission-level performance and system sensitivity.

PROFESSIONAL TRAINING

Supervised Machine Learning (Regression & Classification)

DeepLearning.AI - Andrew Ng

2020 | Coursera

Recherche reproductible : principes méthodologiques pour une science transparente

FUN MOOC – Inria

2025 | France Université Numérique

Software, Data, AI and Systems – completed courses

OpenClassrooms

2024 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, web development, algorithms and software engineering fundamentals.

Selective Graduate Program in Fundamental Physics (Magistère)

Université Paris-Saclay

2016 – 2018 | Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

2017 – 2018 | Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

2015 – 2016 | Orsay, France

PROJECTS

Co-author and full member of the LISA Consortium

ESA/NASA Space Mission - Large-scale International Collaboration

2018 – now

Contribution to software, analysis and validation workflows for the LISA space mission within an international engineering and scientific collaboration involving ESA, NASA and multiple research institutes.
Role: Software workflows, technical coordination, model integration, simulation campaigns, validation activities, complex signal-processing tasks and collaborative engineering support.
Relevance for IVV / embedded systems validation: Work involving complex system behaviour, mission-level performance constraints, simulation-based validation, consistency checks, non-regression logic, technical documentation, anomaly investigation and cross-functional engineering coordination.
Program scale: Major international space mission with an estimated budget of approximately 2 billion EUR over 10 years.

Co-author and full member of Einstein Telescope and LIGO–Virgo–KAGRA Collaborations

International Collaborations

2021 – now

Contribution to collaborative developments in data analysis, signal characterization, software methods and technical investigations within the LIGO–Virgo–KAGRA and Einstein Telescope collaborations.
Role: Development of analysis tools, contribution to technical activities and support to complex software workflows in high-standard collaborative environments.
System impact: Studies on environmental and instrumental noise sources helping identify fundamental design constraints and performance limitations for next-generation gravitational-wave observatories.

STRENGTHS

IVVQ, validation & test automation

IVV · IVVQ · V-Cycle · IVVQ Procedures · Validation Campaigns · Test Automation · Python Test Scripts · Documented Code · Script Execution · Test Result Analysis · Non-Regression Testing · Acceptance Criteria · Test Strategy · Test Reports · Validation Evidence · User Guide Writing · Progress Reporting

Systems, diagnostics & interfaces

Complex Systems · Embedded Systems Context · Critical Systems · Defence Environment Interest · System Integration · Software Integration · Simulation-Based Validation

System-Level Analysis · Anomaly Analysis · First-Level Diagnostics · Root Cause Analysis · Hardware / Software Interfaces · Interface Reliability · Operational Data Exchanges · Protocol Constraints Awareness · Data Consistency · Performance Analysis

Tools, software & coordination

Python · Object-Oriented Programming · Linux · Bash · Git · GitLab · GitHub · Docker · CI/CD · SQL · TypeScript · Angular · Node.js · REST API · FTP · JIRA · Confluence · Zenhub · Agile Tooling · Technical Coordination · Stakeholder Communication · Technical English